SATELLITES, NAVIC, GAGAN AND APPLICATIONS

SATELLITE SERVICE TAXONOMY → ORBIT-SERVICE MATRIX → END-TO-END SEGMENT CHAIN → OBSERVATION-TO-DECISION RAIL → COMMUNICATION RELAY LOOP → NAVIC SYSTEM MAP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 SATELLITE SERVICE TAXONOMY

SATELLITE SERVICE TAXONOMY | EARTH OBSERVATION | SENSED EARTH DATA AND DERIVED PRODUCTS | RELAY THROUGH TRANSPONDERS AND GROUND NETWORKS | IMAGING, SOUNDING, RELAY AND WARNING SUPPORT | POSITION, NAVIGATION AND TIMING SIGNALS | PAYLOAD AND SERVICE DEFINE THE CLASS

EARTH OBSERVATION → sensed Earth data and derived products

COMMUNICATION → relay through transponders and ground networks → METEOROLOGY → imaging, sounding, relay and warning support → NAVIGATION → position, navigation and timing signals

CONCEPT / ARCHITECTURE
- EARTH OBSERVATION → sensed Earth data and derived products
- COMMUNICATION → relay through transponders and ground networks

MECHANISM / APPLICATION
- METEOROLOGY → imaging, sounding, relay and warning support
- NAVIGATION → position, navigation and timing signals

READINESS / STATUS LIMIT
- RULE → payload and service define the class
- EARTH OBSERVATION → sensed Earth data and derived products

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → payload and service define the class.

01

STAGE 01 ORBIT-SERVICE MATRIX

ORBIT-SERVICE MATRIX | CONTINUOUS REGIONAL VIEW AND FIXED GROUND GEOMETRY | DAILY-PERIOD GEOMETRY WITH CHANGING SKY POSITION | POLAR LEO | REPEATED HIGH-INCLINATION EARTH COVERAGE | SUN-SYNCHRONOUS LEO | NEAR-CONSTANT LOCAL-SOLAR-TIME IMAGING | ORBIT ENABLES SERVICE BUT DOES NOT NAME THE PAYLOAD

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
GEO → continuous regional view and fixed ground geometry	POLAR LEO → repeated high-inclination Earth coverage	RULE → orbit enables service but does not name the payload
IGSO / GSO → daily-period geometry with changing sky position	SUN-SYNCHRONOUS LEO → near-constant local-solar-time imaging	GEO → continuous regional view and fixed ground geometry

SYSTEM / DEVICE / INSTITUTION
- GEO → continuous regional view and fixed ground geometry
- IGSO / GSO → daily-period geometry with changing sky position

DECISIVE TECHNICAL DISTINCTION
- POLAR LEO → repeated high-inclination Earth coverage
- SUN-SYNCHRONOUS LEO → near-constant local-solar-time imaging

STATUS / UNIT / EVIDENCE LIMIT
- RULE → orbit enables service but does not name the payload
- GEO → continuous regional view and fixed ground geometry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → orbit enables service but does not name the payload.

02

STAGE 02 END-TO-END SEGMENT CHAIN

END-TO-END SEGMENT CHAIN | SPACE SEGMENT | SATELLITE BUS AND MISSION PAYLOAD | CONTROL SEGMENT | TRACKING, COMMAND AND HEALTH MANAGEMENT | PROCESSING SEGMENT | CORRECTION, CALIBRATION OR DATA PRODUCTS | USER SEGMENT

SPACE SEGMENT → satellite bus and mission payload → CONTROL SEGMENT → tracking, command and health management → PROCESSING SEGMENT → correction, calibration or data products → USER SEGMENT → receiver, department or operational workflow → OUTCOME → useful service only after all segments function

STARTING CONDITION / INPUT
- SPACE SEGMENT → satellite bus and mission payload
- CONTROL SEGMENT → tracking, command and health management

SCIENTIFIC OR ENGINEERING MECHANISM
- PROCESSING SEGMENT → correction, calibration or data products
- USER SEGMENT → receiver, department or operational workflow

OUTPUT / FAILURE BOUNDARY
- OUTCOME → useful service only after all segments function
- SPACE SEGMENT → satellite bus and mission payload

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → useful service only after all segments function.

03

STAGE 03 OBSERVATION-TO-DECISION RAIL

OBSERVATION-TO-DECISION RAIL | RECORDS REFLECTED OR EMITTED ENERGY | SENDS MEASUREMENTS TO A GROUND STATION | CALIBRATION, CORRECTION AND PRODUCT GENERATION | USER AGENCY | INTERPRETS THE PRODUCT | MAPPING, CROP, WATER, WARNING OR RESPONSE ACTION

SENSOR → records reflected or emitted energy → DOWNLINK → sends measurements to a ground station → PROCESSING → calibration, correction and product generation → USER AGENCY → interprets the product → DECISION → mapping, crop, water, warning or response action

STARTING CONDITION / INPUT

SCIENTIFIC OR ENGINEERING MECHANISM

OUTPUT / FAILURE BOUNDARY

- SPACE SEGMENT → satellite bus and mission payload
- CONTROL SEGMENT → tracking, command and health management

- PROCESSING SEGMENT → correction, calibration or data products
- USER SEGMENT → receiver, department or operational workflow

- OUTCOME → useful service only after all segments function
- SPACE SEGMENT → satellite bus and mission payload

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → useful service only after all segments function.

03

STAGE 03

OBSERVATION-TO-DECISION RAIL

OBSERVATION-TO-DECISION RAIL · RECORDS REFLECTED OR EMITTED ENERGY · SENDS MEASUREMENTS TO A GROUND STATION · CALIBRATION, CORRECTION AND PRODUCT GENERATION · USER AGENCY · INTERPRETS THE PRODUCT · MAPPING, CROP, WATER, WARNING OR RESPONSE ACTION

SENSOR → records reflected or emitted energy → DOWNLINK → sends measurements to a ground station → PROCESSING → calibration, correction and product generation → USER AGENCY → interprets the product → DECISION → mapping, crop, water, warning or response action

STARTING CONDITION / INPUT
- SENSOR → records reflected or emitted energy
- DOWNLINK → sends measurements to a ground station

SCIENTIFIC OR ENGINEERING MECHANISM
- PROCESSING → calibration, correction and product generation
- USER AGENCY → interprets the product

OUTPUT / FAILURE BOUNDARY
- DECISION → mapping, crop, water, warning or response action
- SENSOR → records reflected or emitted energy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DECISION → mapping, crop, water, warning or response action.

04

STAGE 04

COMMUNICATION RELAY LOOP

COMMUNICATION RELAY LOOP · TERRESTRIAL STATION SENDS A SIGNAL · RECEIVES, SHIFTS, AMPLIFIES AND RETRANSMITS · SIGNAL REACHES A SERVICE FOOTPRINT · GROUND NETWORK · ROUTES CONTENT TO FINAL USERS · SATELLITE CAPACITY IS NOT THE SAME AS DELIVERED ACCESS

UPLINK → terrestrial station sends a signal → TRANSPONDER → receives, shifts, amplifies and retransmits → DOWNLINK → signal reaches a service footprint → GROUND NETWORK → routes content to final users → RULE → satellite capacity is not the same as delivered access

STARTING CONDITION / INPUT
- UPLINK → terrestrial station sends a signal
- TRANSPONDER → receives, shifts, amplifies and retransmits

SCIENTIFIC OR ENGINEERING MECHANISM
- DOWNLINK → signal reaches a service footprint
- GROUND NETWORK → routes content to final users

OUTPUT / FAILURE BOUNDARY
- RULE → satellite capacity is not the same as delivered access
- UPLINK → terrestrial station sends a signal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → satellite capacity is not the same as delivered access.

05

STAGE 05

NAVIC SYSTEM MAP

NAVIC SYSTEM MAP · REGIONAL CONSTELLATION · INDEPENDENT INDIAN PNT SOURCE · GEO PLUS IGSO DESIGN · NOMINAL REGIONAL GEOMETRY · CONTROL SEGMENT · ORBIT AND CLOCK MONITORING · USER RECEIVER

REGIONAL CONSTELLATION → independent Indian PNT source

GEO PLUS IGSO DESIGN → nominal regional geometry → CONTROL SEGMENT → orbit and clock monitoring → USER RECEIVER → measures signal travel time

CONCEPT / ARCHITECTURE
- REGIONAL CONSTELLATION → independent Indian PNT source
- GEO PLUS IGSO DESIGN → nominal regional geometry

MECHANISM / APPLICATION
- CONTROL SEGMENT → orbit and clock monitoring
- USER RECEIVER → measures signal travel time

READINESS / STATUS LIMIT
- RULE → nominal design is not dated functional health
- REGIONAL CONSTELLATION → independent Indian PNT source

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → nominal design is not dated functional health.

06

STAGE 06

NAVIC SERVICE AND SIGNAL STACK

NAVIC SERVICE AND SIGNAL STACK · OPEN POSITIONING SERVICE · ENCRYPTED SERVICE FOR AUTHORISED USERS · ORIGINAL SIGNAL LAYERS IN OWNER · MASS-MARKET INTEROPERABILITY DIRECTION · PAYLOAD SUPPORT DOES NOT PROVE DEVICE ADOPTION

SPS → open positioning service

RS → encrypted service for authorised users → L5 / S → original signal layers in owner → L1 → mass-market interoperability direction

CONCEPT / ARCHITECTURE
- SPS → open positioning service
- RS → encrypted service for authorised users

MECHANISM / APPLICATION
- L5 / S → original signal layers in owner
- L1 → mass-market interoperability direction

READINESS / STATUS LIMIT
- RULE → payload support does not prove device adoption
- SPS → open positioning service

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → payload support does not prove device adoption.

06 STAGE 06 NAVIC SERVICE AND SIGNAL STACK

NAVIC SERVICE AND SIGNAL STACK · OPEN POSITIONING SERVICE · ENCRYPTED SERVICE FOR AUTHORISED USERS · ORIGINAL SIGNAL LAYERS IN OWNER · MASS-MARKET INTEROPERABILITY DIRECTION · PAYLOAD SUPPORT DOES NOT PROVE DEVICE ADOPTION

SPS → open positioning service

RS → encrypted service for authorised users → L5 / S → original signal layers in owner → L1 → mass-market interoperability direction

CONCEPT / ARCHITECTURE
- SPS → open positioning service
- RS → encrypted service for authorised users

MECHANISM / APPLICATION
- L5 / S → original signal layers in owner
- L1 → mass-market interoperability direction

READINESS / STATUS LIMIT
- RULE → payload support does not prove device adoption
- SPS → open positioning service

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → payload support does not prove device adoption.

07 STAGE 07 CONSTELLATION-HEALTH FIREWALL

CONSTELLATION-HEALTH FIREWALL · CUMULATIVE MISSION COUNT · INTENDED ORBIT · ORBIT-RAISING COMPLETED · SPACECRAFT PERFORMING A STATED FUNCTION · PNT CAPABLE · FULL NAVIGATION SERVICE CONTRIBUTION · MESSAGE BROADCAST

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
LAUNCHED → cumulative mission count	OPERATIONAL → spacecraft performing a stated function	MESSAGE BROADCAST → useful but not equivalent to PNT
INTENDED ORBIT → orbit-raising completed	PNT CAPABLE → full navigation service contribution	LAUNCHED → cumulative mission count

SYSTEM / DEVICE / INSTITUTION
- LAUNCHED → cumulative mission count
- INTENDED ORBIT → orbit-raising completed

DECISIVE TECHNICAL DISTINCTION
- OPERATIONAL → spacecraft performing a stated function
- PNT CAPABLE → full navigation service contribution

STATUS / UNIT / EVIDENCE LIMIT
- MESSAGE BROADCAST → useful but not equivalent to PNT
- LAUNCHED → cumulative mission count

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MESSAGE BROADCAST → useful but not equivalent to PNT.

08 STAGE 08 GAGAN CORRECTION CHAIN

GAGAN CORRECTION CHAIN · REFERENCE STATIONS · MEASURE GNSS ERROR · MASTER CONTROL · COMPUTES CORRECTION AND INTEGRITY · UPLINK STATIONS · SEND AUGMENTATION MESSAGE · GEO PAYLOAD

REFERENCE STATIONS → measure GNSS error → MASTER CONTROL → computes correction and integrity → UPLINK STATIONS → send augmentation message → GEO PAYLOAD → rebroadcasts over a wide area → AIRCRAFT RECEIVER → applies correction and checks integrity

STARTING CONDITION / INPUT
- REFERENCE STATIONS → measure GNSS error
- MASTER CONTROL → computes correction and integrity

SCIENTIFIC OR ENGINEERING MECHANISM
- UPLINK STATIONS → send augmentation message
- GEO PAYLOAD → rebroadcasts over a wide area

OUTPUT / FAILURE BOUNDARY
- AIRCRAFT RECEIVER → applies correction and checks integrity
- REFERENCE STATIONS → measure GNSS error

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AIRCRAFT RECEIVER → applies correction and checks integrity.

09 STAGE 09 ACCURACY-INTEGRITY GATE

ACCURACY-INTEGRITY GATE · ESTIMATED POSITION CLOSE TO TRUE POSITION · WARNING WHEN INFORMATION MUST NOT BE USED · TIME TO ALERT · WARNING MUST ARRIVE WITHIN A SAFE INTERVAL · AVIATION USE · CERTIFICATION GOVERNS OPERATIONAL ACCEPTANCE · A PRECISE SIGNAL WITHOUT INTEGRITY IS UNSAFE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
ACCURACY → estimated position close to true position	TIME TO ALERT → warning must arrive within a safe interval	RULE → a precise signal without integrity is unsafe
INTEGRITY → warning when information must not be used	AVIATION USE → certification governs operational acceptance	MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

SYSTEM / DEVICE / INSTITUTION
- ACCURACY → estimated position close to true position
- INTEGRITY → warning when information must not be used

DECISIVE TECHNICAL DISTINCTION
- TIME TO ALERT → warning must arrive within a safe interval
- AVIATION USE → certification governs operational acceptance

STATUS / UNIT / EVIDENCE LIMIT
- RULE → a precise signal without integrity is unsafe
- MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a precise signal without integrity is unsafe.

A PRECISE SIGNAL WITHOUT INTEGRITY IS UNSAFE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
ACCURACY → estimated position close to true position	TIME TO ALERT → warning must arrive within a safe interval	RULE → a precise signal without integrity is unsafe
INTEGRITY → warning when information must not be used	AVIATION USE → certification governs operational acceptance	MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

SYSTEM / DEVICE / INSTITUTION

- ACCURACY → estimated position close to true position
- INTEGRITY → warning when information must not be used

DECISIVE TECHNICAL DISTINCTION

- TIME TO ALERT → warning must arrive within a safe interval
- AVIATION USE → certification governs operational acceptance

STATUS / UNIT / EVIDENCE LIMIT

- RULE → a precise signal without integrity is unsafe
- MUST REMEMBER: Satellite analysis joins orbit, bus, payload, ground segment and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a precise signal without integrity is unsafe.

10

STAGE 10 SBAS-GBAS INSTITUTION MAP

SBAS-GBAS INSTITUTION MAP · WIDE-AREA CORRECTIONS REBROADCAST FROM SATELLITES · LOCAL AIRPORT CORRECTIONS BROADCAST FROM THE GROUND · ISRO + AAI · GAGAN DEVELOPMENT PARTNERSHIP · AVIATION OPERATIONAL ROLE · AVIATION CERTIFICATION ROLE · CLOSE DISTINCTION

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
SBAS → wide-area corrections rebroadcast from satellites	ISRO + AAI → GAGAN development partnership	DGCA → aviation certification role
GBAS → local airport corrections broadcast from the ground	AAI → aviation operational role	CLOSE DISTINCTION: NavIC is India's regional satellite-navigation system, while GAGAN is...

SYSTEM / DEVICE / INSTITUTION

- SBAS → wide-area corrections rebroadcast from satellites
- GBAS → local airport corrections broadcast from the ground

DECISIVE TECHNICAL DISTINCTION

- ISRO + AAI → GAGAN development partnership
- AAI → aviation operational role

STATUS / UNIT / EVIDENCE LIMIT

- DGCA → aviation certification role
- CLOSE DISTINCTION: NavIC is India's regional satellite-navigation system, while GAGAN is...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: NavIC is India's regional satellite-navigation system, while GAGAN is.

11

STAGE 11 PUBLIC-VALUE ANSWER SPINE

PUBLIC-VALUE ANSWER SPINE · OBSERVATION COMMUNICATION METEOROLOGY PNT OR AUGMENTATION · SPACE GROUND PROCESSING AND USER SEGMENTS · NAVIC CONSTELLATION FROM GAGAN SBAS · STANDARDS ADOPTION INTEGRITY PRIVACY JAMMING AND SPOOFING · DATED HEALTH FREQUENCY COVERAGE ACCURACY AND CERTIFICATION · EVIDENCE LIMIT · STATE CONSTELLATION/SERVICE AREA, ORBIT

CLASSIFY → observation communication meteorology PNT or augmentation → **TRACE → space ground processing and user segments** → **DISTINGUISH → NavIC constellation from GAGAN SBAS** → **TEST → standards adoption integrity privacy jamming and spoofing** → **VERIFY → dated health frequency coverage accuracy and certification**

CONCEPT / ARCHITECTURE

- CLASSIFY → observation communication meteorology PNT or augmentation
- TRACE → space ground processing and user segments

MECHANISM / APPLICATION

- DISTINGUISH → NavIC constellation from GAGAN SBAS
- TEST → standards adoption integrity privacy jamming and spoofing

READINESS / STATUS LIMIT

- VERIFY → dated health frequency coverage accuracy and certification
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State constellation/service area, orbit,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State constellation/service area, orbit,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State constellation/service area, orbit,.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · DEVELOPS AND OPERATES SATELLITE PLATFORMS, THE NAVIC SPACE AND · ISRO BUILDS AND OPERATES THE NAVIGATION CONSTELLATION — IT · AAI + ISRO · JOINT DEVELOPERS OF GAGAN. · AAI IS THE OPERATIONAL OWNER ON THE AVIATION SIDE · USER MINISTRIES/AGENCIES SUCH AS IMD AND OTHER MOES BODIES · CONVERT METEOROLOGICAL SATELLITE INPUTS INTO ACTIONABLE FORECASTS AND WARNINGS.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- ISRO: develops and operates satellite platforms, the NavIC space and control segment and major application systems.
- (Precisely: ISRO builds and operates the navigation constellation — it is not the sole "owner" of every downstream service built on it.)
- AAI + ISRO: joint developers of GAGAN.
- AAI is the operational owner on the aviation side, DGCA is the certifying regulator (RNP 0.1 in 2013,

READINESS, UNIT OR STATUS LIMIT

- User ministries/agencies such as IMD and other MoES bodies: convert meteorological satellite inputs into actionable forecasts and warnings.
- NRSC/Bhuvan ecosystem: exemplifies how data becomes policy utility.
- CAUTION: Navigation systems sit at the overlap of civilian convenience, critical infrastructure and strategic autonomy
- this makes standards, signal policy and device ecosystem choices important policy questions.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- CAUTION: GAGAN's aviation utility shows that "space" is also an air-navigation governance issue, not only a scientific one.
- CAUTION: Broader NavIC adoption depends on procurement standards, handset/chipset support, automotive integration and inter-operability politics.
- CAUTION: An advanced answer can note that data accessibility, security restrictions and
- commercial receiver availability often determine actual impact more than satellite launch success.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels